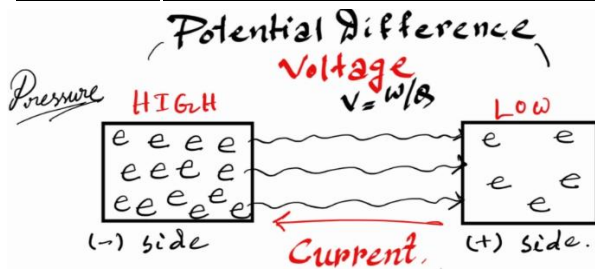


Basics about Current Voltage and Resistor

Quantity	Meaning	Formula	Symbol	Unit
Voltage	Electrical potential difference	$V = W / Q$	V	Volt (V)
Current	Rate of flow of electric charge DC (Direct Current): Flows in one constant direction (batteries, solar cells) AC (Alternating Current): Periodically reverses direction (mains supply, 50/60 Hz)	$I = Q / t$	I	Ampere (A)
Resistance	Opposition to current	Ohm's Law: $V = I \times R$	R	Ohm (Ω)
Power	Rate of electrical energy transfer	$P = I^2 R = V^2 / R = VI$	P	Watt (W)



Key Rules

1. **Voltage** exists between two points even with **no current flowing** (e.g., an open switch — battery still has voltage across terminals).
2. **Current** only flows when there's a **closed/complete circuit**.
3. In a **series circuit**: current is the same everywhere; voltage divides across components.
4. In a **parallel circuit**: current divides among branches; voltage is the same across each branch.
5. **Voltage is measured across (in parallel with)** a component using a voltmeter (high internal resistance).
6. **Current is measured in series** with a component using an ammeter (low internal resistance).

Active and Passive Elements

Passive Elements: Components that cannot generate or amplify energy on their own. They only store, dissipate, or oppose the flow of electrical energy. They cannot provide power gain.

- Examples: Resistor, Capacitor, Inductor

Active Elements: Components capable of delivering energy to a circuit, and/or providing power/voltage/current gain. They require an external power source (DC supply) to operate and can control the flow of current using a small signal.

- Examples: Transistor, Diode (as a rectifying/active nonlinear device), Op-amp, Amplifier circuits

1. Resistor

A resistor is a passive two-terminal component when current flowing through a conductor faces opposition due to **collisions between free electrons** and the **atomic lattice of the material** converting electrical energy into heat. This opposition is called **resistance (R)**

- **Fixed Resistors** – carbon film, metal film, wire-wound
- **Variable Resistors** – potentiometers, rheostats
- **Special** – thermistor (temperature-dependent), LDR (light-dependent)

Important Rules

1. **Series combination:** $R_{\text{total}} = R_1 + R_2 + R_3 + \dots$
2. **Parallel combination:** $1/R_{\text{total}} = 1/R_1 + 1/R_2 + 1/R_3 + \dots$
3. **Color code** determines resistance value and tolerance (4-band/5-band code).
4. Resistance depends on: material (resistivity ρ), length (L), and cross-sectional area (A): $R = \rho L/A$

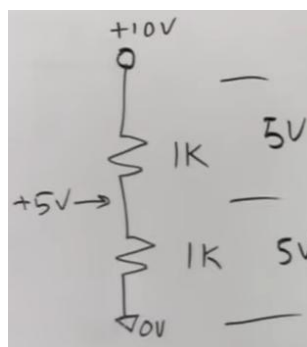
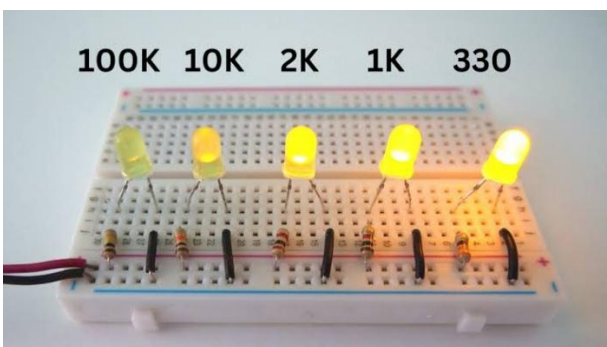
1. Current Limiting (Protecting LEDs, Transistors)

Problem: LEDs and many semiconductor junctions have very low internal resistance. If connected directly to a voltage source, they'd draw excessive current and burn out instantly.

How it works:

- A resistor placed in **series** with the LED limits the current using Ohm's Law.
- Calculation: $R = (V_{\text{supply}} - V_{\text{LED}}) / I_{\text{desired}}$
- Example: 5V supply, LED forward voltage = 2V, desired current = 20mA (0.02A) ; $R = (5 - 2) / 0.02 = 150 \Omega$

The resistor "drops" the excess voltage (3V in this case) and forces the current to stay at the safe, desired value, since current is inversely proportional to resistance at a fixed voltage difference.



2. Voltage Division (Potential Dividers)

Problem: You often need a smaller voltage from a larger supply (e.g., 3.3V from a 5V source) without a separate power supply.

How it works:

- Two resistors (R1, R2) are connected in **series** across the supply voltage. The output is taken from the **junction between them**.
- Since current is the same through both resistors (series circuit), voltage drops proportionally to resistance:

$$V_{out} = V_{in} \times (R2 / (R1 + R2))$$

- Example: $V_{in} = 10V$, $R1 = 1k\Omega$, $R2 = 1k\Omega$; $V_{out} = 10 \times (1000 / (1000 + 1000)) = 5V$
- This works because each resistor "claims" a share of the total voltage proportional to its resistance (KVL: voltage drops must sum to source voltage).

Limitation: Only works well with high-impedance loads at the output — if you draw significant current from the tap point, it loads the divider and changes the output voltage.

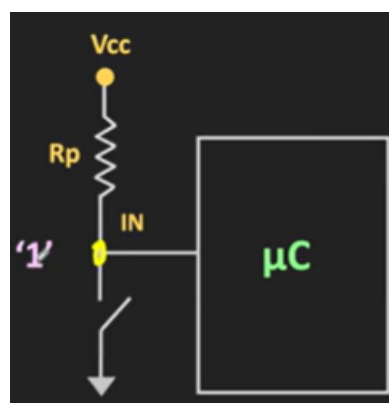
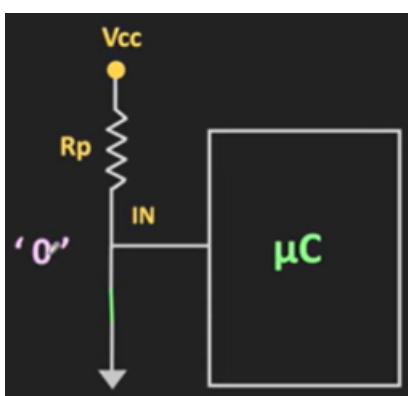
3. Pull-Up / Pull-Down Resistors (Digital Circuits)

Problem: A digital input pin (microcontroller, logic gate) left unconnected ("floating") picks up noise and randomly reads HIGH or LOW — undefined state.

How it works:

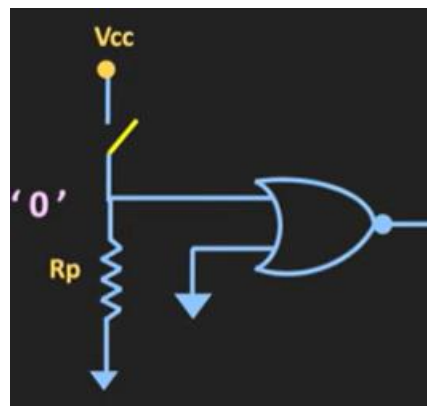
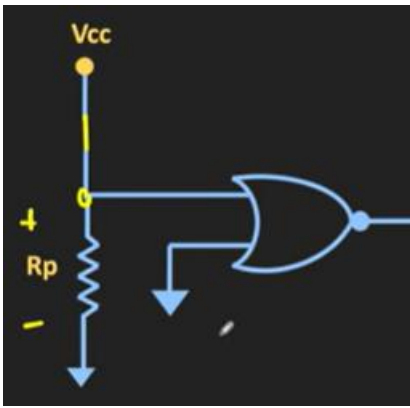
Pull-Up Resistor:

- Connected between the **signal line and V_{supply} (+5V or +3.3V)**.
- When the switch is **open**, the resistor pulls the line HIGH (current flows from supply through resistor to the pin, negligible drop since pin draws ~0 current → **pin reads V_{supply}**).
- When the switch is **closed** (connects line to GND), current flows through the resistor to ground, and the **pin reads LOW (~0V)**.
- Also The resistor value (typically 1kΩ–10kΩ) limits current when the switch is closed, preventing a direct short from supply to ground.



Pull-Down Resistor:

- Same idea, but connected between the **signal line and GND**.
- Default state (switch open) = LOW.
- When switch is closed (connects to supply) = HIGH.



Why resistance value matters:

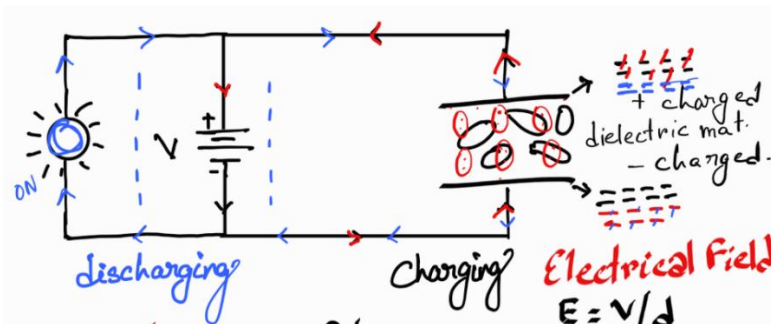
- Too low → wastes power, draws significant current when switch is closed.
- Too high → line becomes susceptible to noise/interference again (slow RC response, weak drive).

2. Capacitor

A capacitor is a passive two-terminal component that **stores electrical energy in the form of an electric field** between two conductive plates separated by an insulating material (dielectric).

Working Principle

- Two parallel conductive plates separated by a **dielectric** (air, ceramic, mica, electrolyte, plastic film).
- When connected to a voltage source, electrons flow onto one plate (making it negative) and are pulled off the other plate (making it positive), but no charge actually crosses the dielectric. Creating an **electric field** across the dielectric.
- This charge separation stores energy. **Charge stored:** $Q = C \times V$; Q = charge (Coulombs), C = Capacitance (Farads), V = voltage across plates
- **Charging/Discharging behavior:** A capacitor doesn't charge or discharge instantly — voltage across it rises/falls exponentially, governed by the **time constant:** $\tau = R \times C$ (time to reach ~63% of final value)



Important Rules

1. Capacitance depends on plate area (A), distance between plates (d), and dielectric material (ϵ): $C = \epsilon A/d$

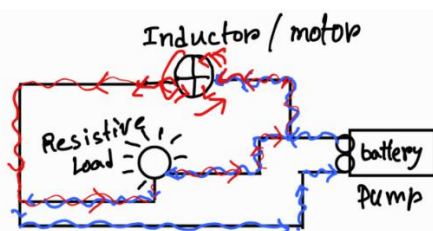
2. **Blocks DC** (once fully charged, no current flows) but **allows AC** to pass (continuously charges/discharges as polarity reverses).
 3. **Voltage across a capacitor cannot change instantaneously** (opposite behavior to inductor's current rule).
 4. **Series combination:** $1/C_{\text{total}} = 1/C_1 + 1/C_2 + \dots$
 5. **Parallel combination:** $C_{\text{total}} = C_1 + C_2 + C_3 + \dots$
 6. **Energy stored:** $E = \frac{1}{2}CV^2$
 7. **Capacitive reactance (opposition to AC):** $X_C = 1/(2\pi fC)$ — **decreases** as frequency increases (opposite of inductor).
 8. Acts as an **open circuit for DC** (steady state) and **low impedance for high-frequency AC**.
-

3. Inductor

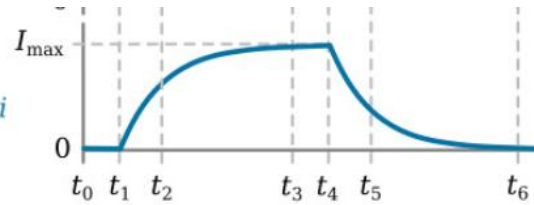
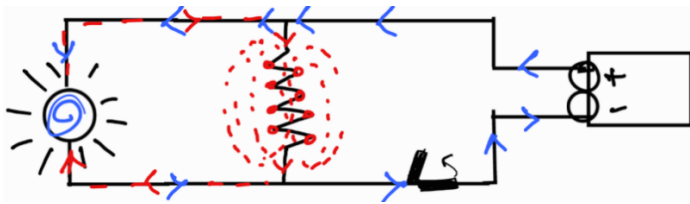
An inductor is a passive two-terminal component, typically a coil of wire, that **stores energy in a magnetic field** when current flows through it, and opposes any *change* in current.

Working Principle

- Wire wound into a coil, often around a core (air, iron, ferrite) to increase inductance.
- Current flow from battery to the bulb as initially the resistance in Inductor is higher. But little amount of e flow through the coil and gradually reduce resistance and create **magnetic field** (right-hand rule).
- If current changes, the magnetic flux through the coil changes, and by **Faraday's Law**, this induces an EMF in the coil itself.
- Inductor doesn't like changes in Current Flow.
- By **Lenz's Law**, this induced EMF **opposes the change** that created it (opposes increase or decrease in current). $V = L \times (di/dt)$
- By pull electron when switch ON and push the electron when switch OFF.
- This is why inductors resist sudden current changes — they act like "electrical inertia" for current.



1. initially motor resistance > resistive load.
2. motor start to rotate. → No resistance
3. pump stop still water flow bcz motor rotate and inertia of water.



Important Rules

1. **Current through an inductor cannot change instantaneously** (opposite of capacitor's voltage rule).
2. **Series:** $L_{\text{total}} = L_1 + L_2 + \dots$
3. **Parallel:** $1/L_{\text{total}} = 1/L_1 + 1/L_2 + \dots$
4. **Energy stored:** $E = \frac{1}{2}LI^2$
5. **Inductive reactance:** $X_L = 2\pi fL$ — **increases** with frequency.
6. Acts as a **short circuit for DC** (steady state, just wire resistance) and **high impedance for high-frequency AC**.

Property	Capacitor	Inductor
Stores energy in	Electric field	Magnetic field
Opposes change in	Voltage	Current
DC behavior (steady state)	Open circuit (blocks)	Short circuit (passes)
AC behavior	Passes (low reactance at high f)	Blocks (high reactance at high f)
Reactance formula	$X_C = 1/(2\pi fC)$ — decreases with f	$X_L = 2\pi fL$ — increases with f
Energy formula	$E = \frac{1}{2}CV^2$	$E = \frac{1}{2}LI^2$
Series formula	$1/C_{\text{total}} = \Sigma(1/C)$	$L_{\text{total}} = \Sigma L$
Parallel formula	$C_{\text{total}} = \Sigma C$	$1/L_{\text{total}} = \Sigma(1/L)$
Instant change allowed in	Current (voltage can't jump)	Voltage (current can't jump)